

SAMEEK BHATTACHARYA

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PROFESSIONAL SUMMARY

AI/ML and Software Development professional with 5 years of experience across software development, research and deployment, skilled in Python, PyTorch, Scikit-learn and deep learning models among others. Ability to architect end-to-end pipelines in adversarial audio attacks, RAG systems, NLP and computer vision, translating research into real-world applications. Actively seeking AI/ML Engineer, Data Scientist and Software Developer roles, leveraging an MS in Artificial Intelligence from the University of North Texas.

WORK EXPERIENCE

University of North Texas

United States

Data Scientist Intern

May 2025 - July 2025

- Architected an end-to-end **data pipeline** using **Python (Pandas, NumPy)** to collect, de-identify, and process student registration and feedback data, applying **NLP preprocessing (NLTK)** to extract quantifiable **sentiment trends** across 6 weekly student groups.
- Conducted end-to-end **EDA** and **statistical analysis** using **Python (Pandas, Matplotlib, Seaborn, SciPy)** on longitudinal **demographic** and **pre/post-assessment data, engineering features** to understand knowledge-gain patterns and delivering an insights report with data-driven curriculum recommendations adopted for the next program cycle.
- Instructed **35+ students** weekly during a **6-week, state-funded AI summer camp**, delivering hands-on training in **Python programming, hardware development (K210 Kit)**, and **robotics (XGO robo-dog with micro:bit via Microsoft MakeCode)**.

Research Assistant (Visual Computing and Biometric Security Lab)

Feb 2025 - Current

- Developed a **GAN-based attack** on **latent space** of audio data for implementing in **real-time** cases and evaluated existing state-of-the-art models, outperforming other white box attack mechanisms. **GAN-based attack** achieved - **Attack Success Rate 99.11%+, attack speed 0.007 sec**.
- Dataset & Tasks:** **ASVspoof2019** - Deepfake detection, **DCASE2019** and **UrbanSound8k** - Sound event classification, **LibriSpeech** - Speaker Verification
- Models Used & Attacks:** Evaluated models - **ECAPA-TDNN, PANN-CNN14, ResNet50, Xception, ResNet50** feature extractor with **Logistic Regression** classifier. Implemented attacks - **Gaussian Noise, FGSM, BIM, PGD, DeepFool, Carlini Wagner**.
- Architectural Pipeline:** Extracted **MFCC** and **STFT** features and saving **NumPy** arrays using **librosa** library from raw **.flac** and **.wav** data. Passed through models for **training** and **evaluation**. Implemented attacks and computed **attack evaluation**. Extracted **audio embeddings** from raw audio for speaker verification. Computed **cosine similarity** scores. Passed through **classifier** for **evaluation**, implemented attack and computed **attack evaluation**.

Teaching Assistant (Spring 2026: Introduction to AI | Fall 2025: Computer Science I)

Aug 2025 - May 2026

- Assisted in developing **course materials** and **led lab sessions** for both the courses - **90+ students** in each class, mentoring students on **C/C++ programming, search algorithms** and **fundamental AI concepts**. Graded assignments with **constructive feedback**.

Institute of Engineering and Management

India

Research Assistant

Jan 2023 - May 2024

- Developed **AI-driven tools** including a **life-cycle management framework** with **result dashboard** and a **benchmarking system** for **tabular and image datasets** using **Power BI, Scikit-learn** and **TensorFlow**.
- Analysed **student performance data** to provide actionable insights to the **education board** and oversaw **academic operations** such as **grade card publication, transcripts, certificates, and graduation ceremonies**.

Tata Consultancy Services

India

System Engineer

Aug 2021 - Nov 2022

- Developed and maintained **Johnson & Johnson's training website** for doctors and staff, working as a **full-stack web developer** using **HTML, CSS, Bootstrap, JavaScript, Git**, and **Atlassian tools** and **Drupal** as the CMS following agile framework.
- Ensured **Johnson & Johnson compliance standards** through certifications in **iQMS, HSE, Data Privacy, and Health & Safety**.

EDUCATION

University of North Texas (TX, United States)

December 2026

Masters(Thesis) - Artificial Intelligence - MS Distinguished Scholar

GPA - 3.87/4.0

University of Engineering and Management (WB, India)

June 2021

Bachelors - Computer Science and Engineering

GPA - 8.59/10.0

SKILLS

Programming Languages: Python, Java, C, HTML, CSS, C++, JavaScript

Data Science & ML: Retrieval-Augmented Generation (RAG), Audio Processing, Deepfake detection, Sound event classification, Image Processing, Computer Vision, Feature Engineering, LLM, Natural Language Processing, Deep Learning(ECAPA, CNNs, ResNet, Xception), traditional ML models, multimodal data processing, Scikit-learn, Transfer learning, Model fine-tuning

Cloud Technologies: Kaggle, GoogleWorkspace(Colab, Slides), GitHub, Atlassian(Bitbucket), Linux

DevOps & Version Control: Anaconda, Git, CI/CD

Frameworks & Libraries: PyTorch, TensorFlow/Keras, Hugging Face, Weights & Biases, Gemini-API, Streamlit, Bootstrap, Jupyter Notebook, VS Code, Atlassian(Jira, Confluence), Regression Testing, A/B Testing, Agile Framework, Copilot, ChatGPT

Visualization Tools: Power BI, Matplotlib, Seaborn

Other: LeetCode, Arduino, Excel, Adobe Illustrator, Adobe Effects, InShot

PUBLICATIONS (International Conference, P:Published, S:Submitted)

S: Bhattacharya, Sameek, et. al. 2026, February. [Exploiting Neural Audio Codec Latents for Adversarial Audio Attacks](#). Submitted at Interspeech 2026

P: Bhattacharya, Sameek, et al. 2021, December. [Finding a baseline machine learning model in Sci-Kit Learn package in terms of accuracy and efficiency](#). In IEEE Pune Section International Conference (PuneCon), pp. 1-7. IEEE. 2021. India. DOI: 10.1109/PuneCon52575.2021.9686536

PROJECT EXPERIENCE

- [RAG-Powered Interactive Document QA System](#) (*Google Gemini API, gemini-2.5-flash, google-genai, python-dotenv, Python*)
 - Built a dual-mode QA pipeline — a simple Q&A flow routing user queries directly to the **Gemini API**, and a **RAG** mode that prepends **user-supplied** document context to the **prompt** before querying, with both modes maintaining full conversation history and generating a **session summary** on exit; the system handles requests sequentially with **structured error handling**.
 - Achieved **context-aware, hallucination-reduced** responses grounded in user-provided documents, with an average **response time** of **5–15 seconds**, and a default **temperature** of **0.7** balancing creativity with accuracy across use cases such as document Q&A, CV-to-job-description matching, and code review.
- [EasyChords – Application for Playing Piano Chords](#) (*Python, Eel, Pygame, NumPy, Pillow, PyInstaller, HTML/CSS/JS, GitHub Copilot*)
 - Architected a hybrid desktop app with a **Python** backend connected to an **HTML-CSS-JS** frontend via **Eel**; the audio engine performs **MIDI-to-frequency** conversion, builds major/minor scales across seven harmonic degrees, and synthesizes multi-harmonic sine waves with a custom ADSR envelope via **NumPy**, while **Pygame** handles **low-latency** chord playback.
 - Delivered a fully **zero-dependency, offline Windows** executable packaged via **PyInstaller** with an embedded custom circular icon, featuring an adjustable sustain control for post-release chord fade-out and instant playback across **all supported scales** — eliminating any internet or installation requirements for end users.
- [Self-Sustaining Wearable System for Solar Panel-Based UV Exposure Monitoring](#) (*ESP32, INA219, UV Sensor, Solar Panel, Arduino C++, Python, scikit-learn, Jupyter Notebook*)
 - Collected **~5,000** simultaneous **readings** from an **ESP32**-based **wearable** integrating an **INA219** power monitor, **UV sensor**, and **solar panel**; data was **normalized, visualized**, and split **70/15/15** for **train/validation/test**, then fed into **Logistic Regression** and **SVM** classifiers to evaluate whether solar power output reliably proxies **UV intensity** for **binary high-UV classification**.
 - **Logistic Regression** achieved the best performance at **97% accuracy**, outperforming **SVM (96%)** and a **baseline model (93%)** across Accuracy, Precision, Recall, and F1 metrics; **solar panel output closely mirrored UV sensor readings**, validating it as a **cost-effective, self-powering** proxy that eliminates the need for a dedicated UV sensor.
- [Lightweight Automated Essay Scoring \(AES\) via Regression and Optimized Feature Selection](#) (*Python, scikit-learn, pandas, NumPy, TF-IDF, PCA, Univariate Feature Selection, Jupyter Notebook*)
 - Built an AES pipeline on **141 essays** (exceeding the statistically required minimum of 113); extracted **1,221 raw TF-IDF features**, reduced to 53 via **Univariate Feature Selection** (top 50 word-count features plus unique word and stopword counts), applied **PCA**, then **trained** and compared **regression** and **classification** models for **ordinal score prediction**.
 - **Gradient Boosting Regressor** achieved a near-perfect **QWK** of **0.9595** and **RMSE** of **0.9264**, significantly outperforming the **KNN classifier (QWK 0.6266)**, validating that essay scoring is best framed as a regression task and that **53 optimized features** are sufficient for **near-human grading** in **low-resource settings**.
- [Reddit Comment Sentiment Classification: TF-IDF Baseline vs. Fine-Tuned RoBERTa](#) (*Python, HuggingFace Transformers, RoBERTa, TF-IDF, scikit-learn, Logistic Regression, Kaggle, Jupyter Notebook*)
 - **Subsampled 30,000** comments from the **Kaggle 1M Reddit** dataset into **balanced** and **imbalanced** distributions; applied **TF-IDF** vectorization for a **Logistic Regression** baseline and fine-tuned **RoBERTa** for **contextual sentiment classification**, enabling a direct comparison between traditional **bag-of-words** and **transformer-based** approaches across **positive, neutral, and negative sentiment classes**.
 - Fine-tuned **RoBERTa** outperformed the **TF-IDF** baseline with **F1-scores** of **0.65 (positive)** and **0.47 (neutral)**; the baseline degraded sharply under class imbalance, with **negative F1** dropping from **0.41 to 0.11**, confirming that **transformer** contextual embeddings are critical for **robust sentiment classification** on **noisy, imbalanced** social media text.
- [CNN Architecture Comparison for Binary Skin Cancer Classification on ISIC Dataset](#) (*Python, TensorFlow, Keras, ResNet50, DenseNet121, VGG16, InceptionV3, CLAHE, OpenCV, Jupyter Notebook*)
 - Conducted a comparative study of four pretrained CNN architectures — **ResNet50, DenseNet121, VGG16, and InceptionV3** — on **ISIC dermoscopic images** for binary **benign/malignant classification**, evaluating the impact of two preprocessing pipelines, **CLAHE** contrast enhancement and **polynomial (x²) feature transformation**, on model robustness and class-wise **F1** performance.
 - **DenseNet121** with **CLAHE** achieved the best-balanced result with **F1-scores** of **0.89** for both classes, outperforming **ResNet50 (0.88/0.89)**, **VGG16 (0.89/0.87)**, and **InceptionV3 (0.78/0.71)**, demonstrating that dense connectivity paired with contrast enhancement significantly improves **feature reuse** and **generalization** for **medical image classification**.
- [Kaggle Spaceship Titanic – Passenger Survival Prediction with Web Deployment](#) (*Python, CatBoost, scikit-learn, Logistic Regression, pandas, Jupyter Notebook, Streamlit, HuggingFace*)
 - Led a team of 4 as **Project Manager**, applying **feature engineering** on mixed **tabular data** and comparing **Logistic Regression (accuracy: 62.22%, F1: 0.6156, AUC-ROC: 0.6748)** against **CatBoost (accuracy: 79.79%, F1: 0.7914, AUC-ROC: 0.8611)**, confirming gradient boosting's superiority for binary passenger survival prediction.
- Implemented **Gaussian Naïve Bayes** and **KNN** from scratch on the **Scikit-learn digit recognition dataset** (1797 grayscale 8×8 images), achieving **86%** and **99%** accuracy (k=3) respectively through **hyperparameter optimization**.

COMMUNITY SERVICE EXPERIENCE

Kolkata Robotic Society

Co-Founder

India

Sept 2017 - May 2020

- Co-founded a **non-profit educational organization** to promote **tech awareness** by teaching **high school students AI, machine learning, robotics, Arduino, and drones**, reaching **150+ participants** through approx. **20 sessions** and a **state-level workshop**.
- Developed **entrepreneurial skills** by engaging with **investors** and **entrepreneurs** to support and grow the organization.